

1. Domain Name System (DNS) Architecture

Port 53 (UDP/TCP)

- **Core Purpose:** Distributed, hierarchical database translating human-friendly domain names (e.g., `aktu.ac.in`) into 32-bit/128-bit IP addresses.
- **Transport Ports:** UDP Port 53 for interactive standard queries ($\leq 512B$); TCP Port 53 for zone transfers and extended replies ($> 512B$).
- **3 Domain Space Divisions:**
 - *Generic Domains (gTLD):* `.com` (commercial), `.edu` (education), `.gov` (govt), `.org` (non-profit), `.net`.
 - *Country Domains (ccTLD):* 2-letter country code: `.in`, `.us`, `.uk`.
 - *Inverse Domains (PTR):* Reverse mapping (IP $\rightarrow$ Hostname) via `in-addr.arpa`.
- **Recursive vs Iterative Resolution:**
 - *Recursive:* Queried server takes full burden to resolve and return final IP to client stub resolver.
 - *Iterative:* Queried server returns best-effort referral IP pointer to next subordinate nameserver.
- **Caching & TTL:** Answers cached locally for `TTL` seconds to reduce internet traversal delay.

2. DNS Resource Records (RR) Anatomy

Format

<Domain_Name, Value, Type, TTL>			
Type	Mapping Target	Practical Example	
A	Hostname $\rightarrow$ IPv4 (32-bit)	<code>aktu.ac.in $\rightarrow$ 14.139.233.10</code>	
AAAA	Hostname $\rightarrow$ IPv6 (128-bit)	<code>aktu.ac.in $\rightarrow$ 2404:6800::</code>	
CNAME	Alias $\rightarrow$ Canonical Name	<code>www.aktu.ac.in $\rightarrow$ aktu.ac.in</code>	
MX	Domain $\rightarrow$ Mail Exchanger	<code>aktu.ac.in $\rightarrow$ mail.aktu.ac.in</code>	
NS	Zone $\rightarrow$ Authoritative Server	<code>aktu.ac.in $\rightarrow$ ns1.aktu.ac.in</code>	
PTR	IP $\rightarrow$ Hostname (Reverse)	<code>10.233.139.14.in-addr.arpa</code>	

3. World Wide Web & HTTP / HTTPS

Port 80 vs 443

- **URL Syntax:** `protocol://host:port/path` (e.g., `http://aktu.ac.in:80/syllabus`).
- **Stateless Nature:** HTTP maintains zero client history between requests. Stateful sessions managed via **Cookies** and **JWT**.
- **HTTP Methods:** `GET` (fetch resource, idempotent), `POST` (submit payload/data), `HEAD` (headers only), `PUT` (create/replace), `DELETE`.
- **Status Families:** 1xx (Info), 2xx (Success: `200 OK`), 3xx (Redirection: `301 Moved`, `304 Not Modified`), 4xx (Client Error: `400 Bad`, `404 Not Found`), 5xx (Server: `500 Error`).
- **Persistent vs Non-Persistent HTTP:**
 - *HTTP/1.0 (Non-Persistent):* 1 TCP connection per object (2 RTT overhead).
 - *HTTP/1.1 (Persistent):* Single TCP connection for all objects (Connection: `keep-alive`) + Pipelining.
- **HTTP vs HTTPS:** HTTPS runs HTTP over **TLS/SSL (Port 443)**; provides AES encryption, HMAC integrity, and CA Certificate authentication.

4. Remote Login & TELNET Protocol

Port 23 (RFC 854)

- **Local vs Remote Login:** Local login interacts directly with local terminal driver; Remote login sends keystrokes over TCP to remote pseudoterminal driver (PTY).
- **Network Virtual Terminal (NVT):** Universal intermediate representation. 7-bit ASCII characters (0–127); 8th bit high for control commands.
- **Commands:** Starts with **IAC (Interpret As Command = 255 / 0xFF)**. `WILL`, `DO`, `WONT`, `DONT` for option negotiation.
- **Modes:** Default (character echo), Character (keystroke per packet), Line (local buffer edited before Enter).
- **Vulnerability:** Cleartext passwords! Completely replaced by **SSH (Port 22)**.

5. Electronic Mail (E-Mail) Architecture & MIME

4 Tiers

- **4 Scenarios:** 1 (Same system), 2 (Direct connected hosts), 3 (Alice remote, Bob local), 4 (Modern: Both connected to independent mail servers).
- **Triad Components:** User Agent (UA - Compose/Read), Message Transfer Agent (MTA - SMTP Push), Message Access Agent (MAA - POP3/IMAP Pull).
- **MIME (RFC 2045):** Overcomes SMTP 7-bit ASCII limitation; encodes multimedia/binary into ASCII. Headers: `MIME-Version`, `Content-Type`, `Content-Transfer-Encoding` (Base64 ratio 3 raw bytes $\rightarrow$ 4 ASCII chars).

6. Simple Mail Transfer Protocol (SMTP)

Port 25 / 587

- Protocol Nature:** Strictly a **PUSH Protocol** executing over TCP Port 25 (server relay) and Port 587 (client submission).
- 3-Phase Operational Cycle:**
 - 1. *Connection:* Server sends 220 → Client sends HELO / EHLO → Server sends 250 OK .
 - 2. *Mail Transaction:*
 - MAIL FROM: <sender@domain> → 250 OK
 - RCPT TO: <recv@domain> → 250 OK
 - DATA → 354 Start mail input → Body text → <CRLF>.<CRLF> → 250 Queued .
 - 3. *Teardown:* Client sends QUIT → Server sends 221 Closing .
- Relays & MX Lookup:** Messages route through intermediate MTAs using DNS MX priority preference records.
- Email Security Gateways (AKTU PYQ):** On-premise appliances (Cisco IronPort, Barracuda, Fortinet FortiMail, Symantec) and Cloud (Proofpoint, Mimecast) for spam filtering, DLP, and antivirus scanning.

7. Mail Access Protocols: POP3 vs IMAP4

PULL Engine

Parameter	POP3 (RFC 1939)	IMAP4 (RFC 3501)
Ports	Port 110 (995 SSL)	Port 143 (993 SSL)
Storage	Local client disk	Central server database
Sync	No multi-device sync	Full real-time synchronization
Modes	Delete Mode & Keep Mode	Continuous server stateful
Flags	Local flags only	\Seen , \Flagged synced
Folders	No server folders	Create/manage server folders
Partial Fetch	Full message download	Header preview & search

8. File Transfer Protocol (FTP RFC 959)

Dual Connections

- Why Dual Connections?** Separation of concerns: control uses 7-bit ASCII lines; data transfers raw bytes without command choking. Allows ABOR mid-transfer.
- Control Connection (Port 21):** Stays open throughout session. 7-bit ASCII commands (USER , PASS , RETR , STOR , QUIT) and 3-digit replies.
- Data Connection (Port 20 / PASV):** Opens dynamically per file transfer / directory list; closes immediately upon completion.
- Active Mode (PORT):** Client sends PORT IP ,p1,p2 . Server initiates data connection FROM Port 20 TO Client. *Fails behind client NAT firewalls!*
- Passive Mode (PASV):** Client sends PASV . Server opens random high port \$P\$ (\$>1023\$) → Client connects TO Server. *Firewall friendly!*
- Port Math:** $\$ \text{Port} = p_1 \times 256 + p_2$. (e.g., \$(16, 32)\$ implies $16 \times 256 + 32 = 4128$).
- Anonymous FTP:** Username anonymous , Password = user email.

9. TFTP vs FTP vs HTTP Comparison

RFC 1350

- TFTP (Trivial FTP):** Lightweight, UDP Port 69. No authentication. Fixed 512-byte blocks with Stop-and-Wait ACK. Block size < 512B signals EOF. Used in PXE network boot and router IOS firmware flashing.

Feature	FTP (RFC 959)	TFTP (RFC 1350)	HTTP (RFC 2616)
Transport	TCP 20 & 21	UDP 69	TCP 80 / 443
Channels	Dual Connections	Connectionless	Single Stream
Auth	Mandatory	None (Zero)	Header / Token
Directory	Full browsing	No listing	URL paths
Footprint	Heavy daemon	Fits in ROM/BIOS	Rich client/cache

10. Network Management & SNMP

UDP 161 & 162

- **Framework:** Manager (client on NMS) polls Agents (servers on routers/switches).
- **Ports:** UDP 161 (Agent receives Get/Set); UDP 162 (Manager receives Traps).
- **Management Triad (Slide 65 Analogy):**
 - *SMI:* Grammar/syntax rules using ASN.1 and BER encoding.
 - *MIB:* Object database tree. Prefix: 1.3.6.1.2.1 (mib-2).
 - *SNMP:* Protocol packet formats and message passing.
- **3 Functions:** Status Polling (Get), Remote Configuration (Set), Emergency Alarms (Trap).

11. Cryptography Foundations & Symmetric Ciphers

DES & AES

- **4 Pillars:** Confidentiality, Integrity, Authentication, Non-repudiation.
- **Symmetric:** Single shared key \$K\$. \$C = E_K(P)\$, \$P = D_K(C)\$. Key distribution challenge: \$N(N-1)/2\$ keys.
- **Classical:** Substitution (Caesar shift \$k=3\$, Monoalphabetic vs Polyalphabetic) vs Transposition (Rail Fence, Columnar).
- **DES:** 64-bit block, 56-bit key, 16 Feistel rounds (Broken).
- **AES:** 128-bit block, 128/192/256-bit key, SPN network (SubBytes, ShiftRows, MixColumns, AddRoundKey). Global gold standard.

12. Data Compression Techniques

Lossless & Lossy

- **Compression Ratio:** $\text{CR} = \frac{\text{Uncompressed}}{\text{Compressed}}$.
- **Lossless (Exact bits):**
 - *Run-Length (RLE):* AAAAABBBCCDA → A5B3C2D1A1 .
 - *Huffman:* Variable-length prefix codes based on character probabilities.
 - *Lempel-Ziv (LZW):* Dynamic dictionary table index pointers (ZIP/PNG).
- **Lossy Audio Masking (Slide 87–88):**
 - *Frequency Masking:* Louder tone drowns nearby soft frequencies.
 - *Temporal Masking:* Loud impulse (hammer) temporarily desensitizes ear.

13. RSA Asymmetric Algorithm & Math

AKTU PYQ

- **Key Selection Steps:**
 1. Choose distinct large primes \$p\$ and \$q\$.
 2. Modulus: \$n = p \times q\$.
 3. Totient: \$\phi(n) = (p - 1)(q - 1)\$.
 4. Public exponent \$e\$: \$1 < e < \phi(n)\$ and \$\gcd(e, \phi(n)) = 1\$.
 5. Private exponent \$d\$: \$d \cdot e \equiv 1 \pmod{\phi(n)} \implies d = e^{-1} \pmod{\phi(n)}\$.
 6. Public Key = \$(e, n)\$, Private Key = \$(d, n)\$.
- **Formulas:** \$C = P^e \bmod n\$ and \$P = C^d \bmod n\$ (where \$P < n\$).
- **Solved Example (Character "F"):**
 - \$p=3, q=11 \implies n=33, \phi(n)=20\$.
 - Choose \$e=3 \implies \gcd(3, 20)=1\$.
 - \$3d \equiv 1 \pmod{20} \implies 3 \times 7 = 21 \implies d=7\$.
 - Plaintext \$P = 5\$ ('F').
 - **Encrypt:** \$C = 5^3 \bmod 33 = 125 \bmod 33 = 26\$.
 - **Decrypt:** \$P = 26^7 \bmod 33 = 5\$ ('F') ✓.

14. Firewalls & 5 Diagnostic Commands

Slides 89–93

- **Firewalls:** Packet Filtering (L3/L4 stateless IP/Port matching), Stateful Inspection (L4 connection state table), Application Proxy (L7 payload inspection).
- **Digital Signatures:** Hash signed with sender's Private Key \$S = E_{\text{PR}_A}(H(M))\$; verified with Public Key.
- **5 Networking Commands:**
 - `ipconfig /all` : Shows IP, MAC, Subnet, Gateway, DNS; `/flushdns` .
 - `nslookup <domain>` : Resolves hostname to IP & queries MX/A records.
 - `tracert <target>` : Incremental IP TTL hops & latency discovery.
 - `ping <target>` : ICMP Echo Request (Type 8) & Reply (Type 0).
 - `netstat -ano` : Displays active TCP links, listening ports & PIDs.